

$$P(\theta) = \frac{\theta^{\alpha-1} (1-\theta)^{\beta-1}}{B(\alpha, \beta)}, \quad \theta \in [0, 1]$$

$$\propto \theta^{\alpha-1} (1-\theta)^{\beta-1}$$

likelihood

$$P(D|\theta) = \theta^s (1-\theta)^{n-s}$$

Posterior update

$$P(\theta|D) \propto P(D|\theta) \cdot P(\theta)$$

$$= \theta^s (1-\theta)^{n-s} \cdot \theta^{\alpha-1} (1-\theta)^{\beta-1}$$

$$= \theta^{\alpha+s-1} (1-\theta)^{\beta+n-s-1}$$

$$\text{Beta}(\alpha, \beta) \xrightarrow[\text{n trials}]{\text{for } s \text{ success}} \text{Beta}(\alpha+s, \beta+n-s)$$

for Beta (2,2)

$$\mu_0 = 0.5$$

for 5, 5, 1, 5:

$$\mu = \frac{\alpha}{\alpha + \beta}$$
$$\sigma^2 = \frac{\alpha \beta}{(\alpha + \beta)^2 (\alpha + \beta + 1)}$$

① Beta (3,2)

② Beta (4,2)

③ Beta (4,3)

④ Beta (5,3)

